



on the health effects of the cell phone.

What adds to the fear factor is the sudden slew of research on mobile-safety. Check out the statistics: using cell phones for more than 10 years can double the risk of brain cancer; using those for more than 15 minutes could lead to headaches, sleeplessness and fatigue; four times more risk of rare ear tumours after 10 years of heavy cell phone use. It's likely to take 30-40 years of mobile use to show up health risks. Most of the theories doing the rounds focus on three issues: the thermal effect of exposure to radiation from cell phones; the genetic effect of radio waves harming body cells and damaging the DNA; and the carcinogenic effects—do cell phones generate the kind of radiation that could conceivably cause cancer?

"But every day we swim in a sea of radiation produced by electrical appliances, power lines, wiring in buildings, and a slew of other technologies that are part of modern life," says N.K. Joshi, physicist with the Bhabha Atomic Research Centre in Mumbai. "From the microwave oven in the kitchen, the clock-radio next to your bed, to the cellular phone you hold to your ear." The concern over cell phones stems from a form of low frequency radiation that's produced when the gadgets communicate with their base station. While high-frequency radiation—found in ultraviolet rays, X-rays and gamma rays—is known to cause cancer at high doses, the risks of this milder form remain unclear. According to a report published in the *Journal of the National Cancer Institute* last year, for radiation to cause cancer it must break chemical bonds in the body. Mobile radiations are millions of times less energetic to be able to do this, claimed the researchers.

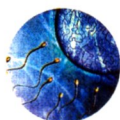
SLEEPLESSNESS

May activate brain's stress system; make people more alert, decrease the ability to wind down and reduce sleep.



STERILITY

May affect male fertility. Long-term users can face lower sperm count, quality and motility.



PREGNANCY

Pregnant women who regularly use mobiles are more likely to have children with behavioural problems.



IRRITABILITY

Greater risk of radiation absorption. Could lead to hyperactivity, behavioural and learning problems in children.



Source: WHO—interphone study, 2008; Sleep, 2008; Fertility & Sterility, 2008; Epidemiology, 2008; Journal of the National Cancer Institute, 2007